

VIT CHENNAI · BMAT201L

Complex Variables & Linear Algebra

Complete Study Guide · 7 Modules · Formula Sheets · Solved Problems

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Analytic Functions of Complex Variables

A function $f(z) = u + iv$ is analytic if it satisfies the Cauchy-Riemann equations — a pair of coupled PDEs ensuring complex differentiability. Analytic functions model potential flow, electrostatics, and are the foundation of conformal mapping.

1.1 Cauchy-Riemann Equations

CAUCHY-RIEMANN (CR) EQUATIONS

CR IN CARTESIAN

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

CR IN POLAR ($Z = RE^{i\theta}$)

$$u_r = \frac{1}{r} v_\theta, \quad \frac{1}{r} u_\theta = -v_r$$

DERIVATIVE

$$f'(z) = u_x + iv_x = v_y - iu_y$$

HARMONIC FUNCTION

$$\nabla^2 u = u_{xx} + u_{yy} = 0$$

HARMONIC CONJUGATE

Given u , find v using CR so that $f = u + iv$ is analytic

ENTIRE FUNCTION

Analytic everywhere (e.g., e^z , $\sin z$, polynomials)

1 Verify analyticity — e^z

Show $f(z) = e^z$ is entire and find $f'(z)$.

SOLUTION

- 1 $e^z = e^x(\cos y + i \sin y) \Rightarrow u = e^x \cos y, v = e^x \sin y$
- 2 $u_x = e^x \cos y = v_y \checkmark$ and $u_y = -e^x \sin y = -v_x \checkmark$ CR satisfied everywhere
- 3 $f'(z) = u_x + iv_x = e^x \cos y + ie^x \sin y = e^z$

$f(z) = e^z$ is entire; $f'(z) = e^z$

2 Find harmonic conjugate

Given $u = x^3 - 3xy^2$, find harmonic conjugate v and $f(z)$.

SOLUTION

- 1 Check: $u_{xx} = 6x$, $u_{yy} = -6x \Rightarrow \nabla^2 u = 0 \checkmark$
- 2 By CR: $v_y = u_x = 3x^2 - 3y^2 \Rightarrow v = 3x^2y - y^3 + g(x)$
- 3 $v_x = 6xy + g'(x) = -u_y = 6xy \Rightarrow g'(x) = 0 \Rightarrow g = c$
- 4 $f(z) = (x^3 - 3xy^2) + i(3x^2y - y^3) = z^3 + ic$

$$v = 3x^2y - y^3 + c; f(z) = z^3$$

1.2 Milne-Thomson Method

MILNE-THOMSON METHOD

To construct $f(z)$ from $u(x,y)$ without finding v explicitly:

$$f(z) = \int [u_x(z, 0) - iu_y(z, 0)]dz + C$$

(Replace $x \rightarrow z$, $y \rightarrow 0$ in the partial derivatives.)

1 Milne-Thomson method

Find $f(z)$ if $u = e^x(x \cos y - y \sin y)$.

SOLUTION

- 1 $u_x = e^x[(x+1) \cos y - y \sin y]$. At $y=0$: $u_x(z, 0) = e^z(z+1)$
- 2 $u_y = e^x(-x \sin y - \sin y - y \cos y)$. At $y=0$: $u_y(z, 0) = 0$
- 3 $f(z) = \int e^z(z+1)dz = \int ze^z dz + \int e^z dz = e^z(z-1) + e^z + C = ze^z + C$

$$f(z) = ze^z + C$$

Module 1 — Key Formulas

CR Cartesian

$$u_x = v_y, \quad u_y = -v_x$$

$f'(z)$

$$u_x + iv_x$$

Harmonic

$$u_{xx} + u_{yy} = 0$$

Complex Integration — Cauchy's Theory

Cauchy's integral theorem and formula are among the most powerful results in mathematics — they evaluate complex integrals purely from function values at isolated points. Taylor and Laurent series classify function behaviour near singularities.

2.1 Cauchy's Theorem & Integral Formula

CAUCHY'S INTEGRAL THEOREM

If $f(z)$ is analytic on and inside a simple closed contour C : $\oint_C f(z) dz = 0$

CAUCHY'S INTEGRAL FORMULA

If $f(z)$ analytic inside C and a inside C :

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz \quad f^{(n)}(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-a)^{n+1}} dz$$

1 Apply CIF

Evaluate $\oint_{|z|=3} \frac{e^z}{z-2} dz$.

SOLUTION

- 1 $f(z)=e^z$ analytic everywhere; $a=2$ inside $|z|=3$ ✓
- 2 By CIF: $= 2\pi i \cdot f(2) = 2\pi i e^2$

$2\pi i e^2$

2 Higher-order CIF

Find $\oint_{|z|=2} \frac{\cos \pi z}{(z-1)^3} dz$.

SOLUTION

- 1 $f(z) = \cos \pi z$, $a=1$ (inside $|z|=2$), $n=2$. Need $f''(1)$.
- 2 $f'(z) = -\pi \sin \pi z$, $f''(z) = -\pi^2 \cos \pi z$. $f''(1) = -\pi^2 \cos \pi = -\pi^2(-1) = \pi^2$
- 3 $\oint = \frac{2\pi i}{2!} f''(1) = \pi i \cdot \pi^2 = \pi^3 i$

$$\pi^3 i$$

2.2 Taylor & Laurent Series

SERIES & SINGULARITY TYPES

TAYLOR (ANALYTIC REGION)

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

LAURENT (ANNULAR REGION)

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n}$$

REMOVABLE SINGULARITY

All $b_n=0$; limit exists and is finite

POLE OF ORDER M

$b_m \neq 0$, $b_n = 0$ for $n > m$; finite principal part

ESSENTIAL SINGULARITY

Infinitely many $b_n \neq 0$

SIMPLE POLE (M=1)

Only $b_1 \neq 0$

1 Laurent expansion

Expand $f(z) = \frac{1}{z(z-1)}$ for $0 < |z| < 1$.

SOLUTION

- 1 Partial fractions: $\frac{-1}{z} + \frac{1}{z-1}$
- 2 For $|z| < 1$: $\frac{1}{z-1} = \frac{-1}{1-z} = -\sum_{n=0}^{\infty} z^n$
- 3 $f(z) = \frac{-1}{z} - \sum_{n=0}^{\infty} z^n = \frac{-1}{z} - 1 - z - z^2 - \dots$

$$f(z) = -1/z - 1 - z - z^2 - \dots \text{ (simple pole at } z=0)$$

Module 2 — Key Formulas

Cauchy's Theorem

$$\oint_C f \, dz = 0 \text{ (analytic)}$$

CIF

$$f(a) = \frac{1}{2\pi i} \oint \frac{f(z)}{z-a} dz$$

nth order CIF

$$\frac{n!}{2\pi i} \oint \frac{f(z)}{(z-a)^{n+1}} dz$$

Residues & Applications to Real Integrals

The Residue Theorem converts complex contour integrals into a simple sum of residues at poles. This powerful technique solves otherwise impossible real definite integrals, evaluating integrals of rational, trigonometric, and improper functions.

3.1 Residue Theorem

RESIDUE FORMULAS

RESIDUE THEOREM

$$\oint_C f(z) dz = 2\pi i \sum_k \text{Res}(f, z_k)$$

RESIDUE AT SIMPLE POLE $z=a$

$$\text{Res}(f, a) = \lim_{z \rightarrow a} (z - a) f(z)$$

RESIDUE AT POLE ORDER m

$$\text{Res}(f, a) = \frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} [(z-a)^m f(z)]$$

1 Evaluate by residue theorem

Evaluate $\oint_{|z|=3} \frac{z^2}{(z+1)^2(z-2)} dz.$

SOLUTION

1 Poles: $z=-1$ (order 2, inside) and $z=2$ (simple, inside)

2 Res at $z=2$: $\lim_{z \rightarrow 2} (z-2) \cdot \frac{z^2}{(z+1)^2(z-2)} = \frac{4}{9}$

3 Res at $z=-1$ (order 2): $\frac{d}{dz} \left[\frac{z^2}{z-2} \right]_{z=-1} = \frac{z^2-4z}{(z-2)^2} \Big|_{z=-1} = \frac{1+4}{9} = \frac{5}{9}$

4 Total: $2\pi i \left(\frac{4}{9} + \frac{5}{9} \right) = 2\pi i$

$$2\pi i$$

2 Real integral by contour integration

Evaluate $\int_0^{2\pi} \frac{d\theta}{5 - 3 \cos \theta}$.

SOLUTION

1 Let $z=e^{i\theta}$, $d\theta=dz/(iz)$, $\cos\theta=(z+z^{-1})/2$

2 Integral becomes $\oint_{|z|=1} \frac{2 dz}{i(3z^2-10z+3)} = \oint \frac{2 dz}{3i(z-3)(z-1/3)}$

3 Only $z=1/3$ inside $|z|=1$. Res at $1/3$: $\frac{2}{3i(1/3-3)} = \frac{2}{3i(-8/3)} = \frac{1}{4i}$

4 $I = 2\pi i \cdot \frac{1}{4i} = \frac{\pi}{2}$

$\pi/2$

Module 3 — Key Formulas

Residue Theorem

$$2\pi i \sum \text{Res}(f, z_k)$$

Simple pole Res

$$\lim_{z \rightarrow a} (z - a)f(z)$$

Trig. integrals

$$z=e^{i\theta}, d\theta=dz/(iz)$$

Conformal mappings preserve angles locally and transform complex domains into simpler shapes where boundary value problems become tractable. The bilinear (Möbius) transformation is the most important class, mapping circles and lines to circles and lines.

4.1 Bilinear (Möbius) Transformation

BILINEAR TRANSFORMATION

GENERAL FORM

$$w = \frac{az + b}{cz + d}, \quad ad - bc \neq 0$$

FIXED POINTS

$$\text{Solve } cz^2 + (d - a)z - b = 0$$

CROSS-RATIO (INVARIANT)

$$\frac{(w - w_1)(w_2 - w_3)}{(w - w_3)(w_2 - w_1)} = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}$$

1 Find bilinear transformation from 3 points

Find bilinear map sending $z_1 = 1, z_2 = i, z_3 = -1$ to $w_1 = 0, w_2 = 1, w_3 = \infty$.

SOLUTION

1 Since $w_3 = \infty$, the map is $w = \frac{z - z_1}{z - z_3} \cdot \frac{z_2 - z_3}{z_2 - z_1}$

2 $w = \frac{z-1}{z+1} \cdot \frac{i+1}{i-1}$. Now $\frac{i+1}{i-1} = \frac{(i+1)(-i-1)}{2} = -i$

3 $w = \frac{-i(z-1)}{z+1}$

$$w = -i(z-1)/(z+1)$$

2 Fixed points

Find fixed points of $w = \frac{6z - 9}{z}$.

SOLUTION

1 Set $w=z$: $z^2 = 6z - 9 \Rightarrow (z - 3)^2 = 0 \Rightarrow z = 3$ (repeated — parabolic type)

Fixed point $z = 3$ (repeated)

4.2 Standard Conformal Mappings

STANDARD MAPPINGS

$W = Z^2$ (SQUARING)

Upper half z -plane $\rightarrow$ full w -plane; doubles angles at origin

$W = E^z$

Strip $0 < y < \pi \rightarrow$ upper half w -plane

$W = Z + 1/Z$ (JOUKOWSKI)

Circles $\rightarrow$ airfoil profiles; aerodynamics applications

Module 4 — Key Formulas

Bilinear

$$w = (az + b)/(cz + d)$$

Fixed points

$$cz^2 + (d - a)z - b = 0$$

Cross-ratio

Invariant under bilinear maps

Vector Spaces & Linear Transformations

Vector spaces provide the abstract framework for linear algebra. Matrices, polynomials, and functions are all vector spaces. Linear transformations preserve the space's structure, with rank and nullity measuring their fundamental properties.

5.1 Vector Spaces, Subspaces & Basis

KEY DEFINITIONS

SUBSPACE TEST (3 CONDITIONS)

Non-empty; closed under addition; closed under scalar multiplication

LINEAR INDEPENDENCE

$$c_1v_1 + \dots + c_nv_n = 0 \Rightarrow \text{all } c_i = 0$$

BASIS

Linearly independent + spans V

DIMENSION

Number of vectors in any basis of V

RANK-NULLITY THEOREM

$$\text{rank}(T) + \text{nullity}(T) = \dim(\text{domain})$$

BASIS TEST VIA DET

$\det \neq 0 \leftrightarrow$ linearly independent (square matrix)

1 Check if set forms a basis

Is $\{(1, 0, 1), (0, 1, 0), (1, 1, 0)\}$ a basis for $\mathbb{R}^3$?

SOLUTION

1 $\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} = 1(0 - 0) - 0 + 1(0 - 1) = -1 \neq 0$

2 $\det \neq 0 \rightarrow$ linearly independent + $n=3$ vectors for $\mathbb{R}^3 \rightarrow$ it is a basis $\checkmark$

Yes, forms a basis for $\mathbb{R}^3$

2 Find rank and nullity

Find rank and nullity of $A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 5 \\ 1 & 2 & 2 & 4 \end{pmatrix}$.

SOLUTION

1 $R_2 \leftarrow R_2 - 2R_1; R_3 \leftarrow R_3 - R_1: \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -1 \\ 0 & 0 & 3 & 1 \end{pmatrix}$

2 $R_3 \leftarrow R_3 - R_2: \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -1 \\ 0 & 0 & 0 & 2 \end{pmatrix}$. Three pivots $\rightarrow$ Rank = 3

3 Nullity = $4 - 3 = 1$ (by Rank-Nullity Theorem)

Rank = 3, Nullity = 1

5.2 Linear Transformations

LINEAR TRANSFORMATIONS

DEFINITION

$$T(u+v) = T(u) + T(v) \text{ and } T(cu) = cT(u)$$

KERNEL (NULL SPACE)

$$\ker(T) = \{v : T(v) = \mathbf{0}\}$$

IMAGE (RANGE)

$$\text{Im}(T) = \{T(v) : v \in V\}$$

1 Matrix of linear transformation

$T(x,y) = (x+2y, 3x-y)$. Find matrix $[T]$ and check invertibility.

SOLUTION

1 $T(e_1) = T(1,0) = (1,3); T(e_2) = T(0,1) = (2,-1)$. Matrix: $[T] = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$

2 $\det = -1 - 6 = -7 \neq 0 \rightarrow$ invertible $\checkmark$

Matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; invertible ($\det = -7$)

Module 5 — Key Formulas

Rank-Nullity

$$\text{rank} + \text{nullity} = \dim(\text{domain})$$

Basis test

$$LI + \text{spans } V$$

Kernel

$$\{v : T(v) = 0\}$$

Inner Products, Eigenvalues & Cayley-Hamilton

Inner product spaces generalise geometry to abstract spaces, enabling orthogonality and projection. Eigenvalues reveal the most fundamental behaviour of transformations; Cayley-Hamilton is a powerful algebraic identity with many computational applications.

6.1 Gram-Schmidt Orthogonalisation

GRAM-SCHMIDT PROCESS

PROJECTION

$$\text{proj}_u v = \frac{\langle v, u \rangle}{\|u\|^2} u$$

STEP 1

$$e_1 = v_1$$

STEP K

$$e_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, e_j \rangle}{\|e_j\|^2} e_j$$

1 Gram-Schmidt

Apply Gram-Schmidt to $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$.

SOLUTION

- 1 $e_1 = (1, 1, 0)$; $\|e_1\|^2 = 2$
- 2 $\langle v_2, e_1 \rangle = 1 + 0 + 0 = 1$. $e_2 = v_2 - \frac{1}{2}e_1 = (1, 0, 1) - (1/2, 1/2, 0) = (1/2, -1/2, 1)$
- 3 Check orthogonality: $\langle e_1, e_2 \rangle = 1/2 - 1/2 + 0 = 0 \checkmark$

$e_1 = (1, 1, 0)$; $e_2 = (1/2, -1/2, 1)$ (orthogonal pair)

6.2 Eigenvalues & Cayley-Hamilton Theorem

EIGENVALUE THEORY

CHARACTERISTIC EQUATION

$$\det(A - \lambda I) = 0$$

EIGENVECTOR

Solve $(A - \lambda I)v = 0$ for each λ

$$\text{TR}(A) = \sum \lambda$$

Sum of eigenvalues = trace

$$\text{DET}(A) = \prod \lambda$$

Product of eigenvalues = determinant

DIAGONALISATION

$A = PDP^{-1}$ where P = eigenvectors, D = $\text{diag}(\lambda)$

CAYLEY-HAMILTON

Every matrix satisfies its characteristic equation: $\phi(A) = 0$

1 Eigenvalues and eigenvectors

Find eigenvalues and eigenvectors of $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.

SOLUTION

1 Char. eq.: $\det \begin{pmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{pmatrix} = (3-\lambda)^2 - 1 = 0 \Rightarrow \lambda^2 - 6\lambda + 8 = 0 \Rightarrow (\lambda - 2)(\lambda - 4) = 0$

2 $\lambda_1 = 2$: $(A - 2I)v = 0 \Rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} v = 0 \Rightarrow v_1 = (1, -1)^T$

3 $\lambda_2 = 4$: $(A - 4I)v = 0 \Rightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0 \Rightarrow v_2 = (1, 1)^T$

$\lambda = 2$: $v = (1, -1)$; $\lambda = 4$: $v = (1, 1)$

2 Cayley-Hamilton — find A^{-1}

Verify Cayley-Hamilton for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and find A^{-1} .

SOLUTION

1 Char. eq.: $\lambda^2 - 5\lambda - 2 = 0$. By C-H: $A^2 - 5A - 2I = 0$

2 $A^2 = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}$. Check: $\begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix} - \begin{pmatrix} 5 & 10 \\ 15 & 20 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \checkmark$

3 From C-H $\times A^{-1}$: $A - 5I - 2A^{-1} = 0 \Rightarrow A^{-1} = \frac{A - 5I}{2} = \frac{1}{2} \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix}$

$$A^{-1} = (1/2)[[-4, 2], [3, -1]]$$

Module 6 — Key Formulas

Char. Equation

$$\det(A - \lambda I) = 0$$

Cayley-Hamilton

$$\phi(A) = O$$

Diagonalisation

$$A = PDP^{-1}$$

Quadratic Forms & Matrix Decompositions

Quadratic forms classify conic sections and surfaces. Positive definiteness determines stability of equilibria. LU decomposition solves linear systems efficiently. Orthogonal diagonalisation (spectral theorem) is the foundation of principal component analysis (PCA).

7.1 Quadratic Forms & Positive Definiteness

QUADRATIC FORMS

QUADRATIC FORM Q

$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ where A is symmetric

POSITIVE DEFINITE (PD)

All eigenvalues >0 ; or all leading principal minors >0

POSITIVE SEMI-DEFINITE

All eigenvalues ≥ 0

NEGATIVE DEFINITE

All eigenvalues <0

INDEFINITE

Mixed signs of eigenvalues

SYLVESTER'S CRITERION (PD)

All leading principal minors $\Delta_1, \Delta_2, \dots >0$

1 Classify a quadratic form

Determine if $Q = 2x^2 + 4xy + 3y^2$ is positive definite.

SOLUTION

- 1 Matrix: $A = \begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}$ (off-diagonal = coefficient/2)
- 2 Leading minor $\Delta_1 = 2 > 0$. $\Delta_2 = \det(A) = 6 - 4 = 2 > 0$.
- 3 Both minors positive $\rightarrow Q$ is **positive definite**

Q is positive definite (Sylvester's criterion satisfied)

7.2 LU Decomposition

LU DECOMPOSITION

GOAL

Factor $A = LU$ where L = lower triangular (1s on diagonal), U = upper triangular. Then solve $Ly=b$ (forward sub) and $Ux=y$ (back sub)

EFFICIENCY

Once $A=LU$ is computed, multiple RHS vectors b can be solved with just $O(n^2)$ operations vs $O(n^3)$ for Gaussian elimination each time

1 LU Decomposition

Find LU decomposition of $A = \begin{pmatrix} 2 & 4 & -2 \\ 1 & 3 & 1 \\ -1 & 2 & 6 \end{pmatrix}$.

STEP-BY-STEP SOLUTION

1 $R2 \leftarrow R2 - (1/2)R1$; $R3 \leftarrow R3 + (1/2)R1$. Multipliers: $\ell_{21} = 1/2$, $\ell_{31} = -1/2$

2 After step 1: $\begin{pmatrix} 2 & 4 & -2 \\ 0 & 1 & 2 \\ 0 & 4 & 5 \end{pmatrix}$

3 $R3 \leftarrow R3 - 4R2$. Multiplier $\ell_{32} = 4$. $U = \begin{pmatrix} 2 & 4 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{pmatrix}$

4 $L = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -1/2 & 4 & 1 \end{pmatrix}$

$L = [[1,0,0],[1/2,1,0],[-1/2,4,1]]$; $U = [[2,4,-2],[0,1,2],[0,0,-3]]$

7.3 Spectral Theorem & Orthogonal Diagonalisation

SPECTRAL THEOREM (SYMMETRIC MATRICES)

Every real symmetric matrix A has:

- All real eigenvalues
- Eigenvectors from different eigenvalues are orthogonal

- $A = QDQ^T$ where Q is orthogonal ($Q^T = Q^{-1}$) and D = diagonal eigenvalue matrix

1 Orthogonal diagonalisation

Orthogonally diagonalise $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.

SOLUTION

1 From Module 6: $\lambda_1=2$ with $v_1=(1,-1)^T$; $\lambda_2=4$ with $v_2=(1,1)^T$. Orthogonal ✓ (symmetric A)

2 Normalise: $\hat{v}_1 = \frac{1}{\sqrt{2}}(1, -1)^T$, $\hat{v}_2 = \frac{1}{\sqrt{2}}(1, 1)^T$

3 $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$. Then $A=QDQ^T$

$A = QDQ^T$ with Q orthogonal, $D = \text{diag}(2, 4)$

Module 7 — Key Formulas

PD test (Sylvester)

All leading principal minors >0

LU: $A=LU$

L lower triangular, U upper triangular

Spectral theorem

Symmetric A : $A=QDQ^T$

BMAT201L — Master Formula Sheet

Modules 1–3: Complex Variables

CR EQUATIONS

$$u_x = v_y, \quad u_y = -v_x$$

$F'(z)$

$$u_x + iv_x$$

CAUCHY'S THEOREM

$$\oint_C f \, dz = 0 \text{ (analytic)}$$

CIF

$$f(a) = \frac{1}{2\pi i} \oint \frac{f(z)}{z-a} dz$$

RESIDUE (SIMPLE POLE)

$$\lim_{z \rightarrow a} (z-a)f(z)$$

RESIDUE THEOREM

$$\oint = 2\pi i \sum \text{Res}$$

Modules 4–5: Mapping & Vector Spaces

BILINEAR

$$w = (az + b)/(cz + d)$$

CROSS-RATIO

Invariant under bilinear maps

RANK-NULLITY

$$\text{rank} + \text{nullity} = \dim(\text{domain})$$

SUBSPACE TEST

Non-empty; closed under + and $\times$

KERNEL

$$\{v : T(v) = 0\}$$

BASIS TEST

LI + spans V

Modules 6–7: Eigenvalues & Forms

CHAR. EQUATION

$$\det(A - \lambda I) = 0$$

CAYLEY-HAMILTON

$$\phi(A) = O$$

DIAGONALISATION

$$A = PDP^{-1}$$

PD (SYLVESTER)

All leading minors >0

SPECTRAL THEOREM

Symmetric: $A = QDQ^T$

GRAM-SCHMIDT

$$e_k = v_k - \sum \text{proj}_{e_j} v_k$$

BMAT201L — Complete

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